ChatGPT

Executive Summary

Key Findings and Metrics

- **Core Calculations:** The tool will compute key metrics like **Revenue Gap** (Approved ARR minus actual revenue), power purchase cost variances, and AT&C losses. For example, if the approved Aggregate Revenue Requirement (ARR) is ₹100 Cr and actual revenue ₹90 Cr, the gap is ₹10 Cr. These metrics align with past KSERC truing-up data.
- **Typical Values:** Historical orders show licensee adjustments on the order of tens of lakhs. For instance, *M/s Infopark (FY2020-21)* had an approved **deficit** of about ₹60.80 L [1], while *Kanan Devan (FY2021-22)* received a **surplus** of ₹58.72 L [2]. These benchmarks indicate the scale of data the AI must handle.
- **Trend Validation:** The tool will also flag anomalies (e.g. cost spikes vs. load growth) by comparing year-over-year trends. Known KSERC cases often involve verifying items like distribution loss, power purchase, and employee costs against approved norms.

Historical Outcomes (Extreme Cases)

- **Technopark (2020-21):** Approved **surplus of ₹280.42 L** [3] (highest observed).
- **CSEZA (2022-23):** Approved **surplus of ₹135.14 L** [4].
- **Kanan Devan (2022-23):** Approved **surplus of ₹116.91 L** [5].
- **Kanan Devan (2021-22):** Approved **surplus of ₹58.72 L** [2].
- **Infopark (2020-21):** Approved **deficit of ₹60.80 L** [1] (licensee had asked for ₹76.53 L).
- **Infopark (2024-25):** Approved **deficit of ₹64.72 L** [6] (licensee had claimed a gap of ₹79.76 L).
- **Reversed Outcomes:** In some cases KSERC flipped the sign of the result. For example, CSEZA's FY2023-24 petition asked for a slight deficit (–₹3.42 L) but the Commission approved a **surplus of ₹87.85 L** (cumulative) [7].

These data points set the bounds for the AI tool's outputs: one should expect surpluses or deficits within these historical ranges, and any values far outside them would be unusual.

Recommended AI Architecture

- **Retrieval-Augmented Generation (RAG):** We recommend a hybrid AI using RAG, where a language model (LLM) is fed with relevant regulatory documents and past orders. RAG ensures that model outputs are grounded in actual sources [8]. For example, when justifying an expense, the system can cite the specific clause or past order.
- **Agent Orchestration:** A LangChain-like agent framework can orchestrate tasks: parse uploaded filings, retrieve regulatory provisions, compute metrics, and produce a reasoning report. This modular workflow is state-of-the-art for compliance tasks [9]. RAG/LangChain approaches have been proven in regulated domains (e.g. pharmaceutical compliance) to improve accuracy and traceability [8] [9].
- **Transparency and Audit:** The architecture will log each decision and its rationale. Since RAG ties answers to source text, KSERC staff can audit the AI's logic step-by-step.

Conclusion

The research establishes how to replicate KSERC's truing-up analysis in software. The tool will compute revenue gaps, validate costs, and detect anomalies using the above metrics. Historical KSERC orders provide benchmarks (e.g. ₹280 L max surplus, ~₹–60 L max deficit) for sanity checks. A retrieval-based LLM system (RAG) is recommended to emulate the regulator's reasoning with full traceability [8] [9]. This approach will yield a robust "AI brain" capable of reviewing submissions at the fidelity of KSERC's own staff.

Sources: Official KSERC orders (see citations) and recent AI compliance literature [1] [3] [4] [5] [2] [8] [9].

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